

SHORT PROOF OF THE BERNOULLI CONJECTURE

ABSTRACT. We give a short proof of the Bernoulli conjecture. The proof was produced by an internal model at OpenAI.

1. INTRODUCTION

Let $\varepsilon = (\varepsilon_i)_{i \in I}$ have independent uniform sign coordinates, and let $G = (g_i)_{i \in I}$ have independent standard Gaussian coordinates. For a nonempty $T \subset \ell^2(I)$, write

$$b(T) = \mathbb{E} \sup_{t \in T} \sum_{i \in I} t_i \varepsilon_i, \quad g(T) = \mathbb{E} \sup_{t \in T} \sum_{i \in I} t_i g_i,$$

with the usual finite-subset interpretation. Recall that $X \preceq_{\text{cx}} Y$ means $\mathbb{E}\Phi(X) \leq \mathbb{E}\Phi(Y)$ for every convex Φ for which the expectations are defined.

Theorem 1.1 (Bernoulli conjecture). *There is a universal constant C such that, whenever $\emptyset \neq T \subset \ell^2(I)$ and $b(T) < \infty$, there exist $U \subset \ell^1(I)$ and $V \subset \ell^2(I)$ such that*

$$T \subset U + V, \quad \sup_{u \in U} \|u\|_1 \leq Cb(T), \quad g(V) \leq Cb(T).$$

The conjecture was proposed by Talagrand [8] and first proved by Bednorz and Latała [2]. Our proof does not invoke their theorem. Its main input is the subgaussian convex-comparison theorem of van Handel [9]. The key input for Theorem 1.1 is the following convex domination result.

Theorem 1.2 (Gaussian-to-Bernoulli comparison). *There is a universal constant C_0 such that, for every $G \sim N(0, I_n)$ and every uniform $\varepsilon \in \{-1, 1\}^n$,*

$$X \preceq_{\text{cx}} G, \quad \|X\|_\infty \leq 1 \quad \text{almost surely} \implies X \preceq_{\text{cx}} C_0 \varepsilon.$$

We use van Handel's Gaussian comparison in the following form: for a universal C_G , if Y is centered and

$$\log \mathbb{E} e^{\langle a, Y \rangle} \leq \frac{K^2}{2} \|a\|_2^2 \quad (a \in \mathbb{R}^n), \tag{1}$$

then $Y \preceq_{\text{cx}} C_G K G'$, where $G' \sim N(0, I_n)$ [9]. By Strassen's theorem [7], this is equivalent to the existence of a martingale coupling with

$$Y = C_G K \mathbb{E}[G' \mid Y]. \tag{2}$$

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Note on concurrent work: During the preparation of this manuscript, we learned of an independent alternate proof of the Bernoulli conjecture by Liu and Zadik [5].

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2. REDUCTION TO THEOREM 1.2

We first give the reduction from Theorem 1.1 to Theorem 1.2.

Proof. By the standard finite-section and compactness reduction, it suffices to prove Theorem 1.1 for a finite $T \subset \mathbb{R}^n$. Set

$$D(T) = \inf_{(v_t)_{t \in T}} \left\{ \max_{t \in T} \|t - v_t\|_1 + \mathbb{E} \max_{t \in T} \langle v_t, G \rangle \right\}. \quad (3)$$

For a measurable probability kernel $q = (q_t)_{t \in T}$, put

$$q_t \geq 0, \quad \sum_{t \in T} q_t = 1, \quad m_t = \mathbb{E}[Gq_t(G)].$$

We claim that

$$D(T) = \sup_q \left\{ \sum_{t \in T} \langle t, m_t \rangle : \sum_{t \in T} \|m_t\|_\infty \leq 1 \right\}. \quad (4)$$

Indeed,

$$\begin{aligned} \max_t \|t - v_t\|_1 &= \sup_{\sum_t \|z_t\|_\infty \leq 1} \sum_t \langle t - v_t, z_t \rangle, \\ \mathbb{E} \max_t \langle v_t, G \rangle &= \sup_q \sum_t \langle v_t, m_t \rangle. \end{aligned}$$

The kernels form a weakly compact convex subset of $L^2(\gamma_n)^T$, so one application of minimax gives

$$D(T) = \sup_{\substack{\sum_t \|z_t\|_\infty \leq 1 \\ q_t \geq 0, \sum_t q_t = 1}} \inf_{(v_t)} \left\{ \sum_t \langle t, z_t \rangle + \sum_t \langle v_t, m_t - z_t \rangle \right\}.$$

The inner infimum is finite only when $z_t = m_t$ for every t : if $m_s \neq z_s$, take $v_s = -R(m_s - z_s)$ and $v_t = 0$ for $t \neq s$ and let $R \rightarrow \infty$. In the remaining case the inner expression is $\sum_t \langle t, m_t \rangle$. This proves (4) and, in particular, explains precisely why the dual variable is forced to equal the Gaussian barycenter.

Fix an admissible kernel in (4), and write $p_t = \mathbb{E}q_t(G)$ and $a_t = \|m_t\|_\infty$. Since $\sum_t (a_t - p_t)_+ \leq \sum_t a_t \leq 1$, choose a probability vector $r = (r_t)_{t \in T}$ such that $r_t \geq (a_t - p_t)_+$. Toss an independent fair coin. On heads, sample a label J from $q(G)$; on tails, sample J independently of G with distribution r . Then

$$\mathbb{P}(J = t) = \frac{p_t + r_t}{2}, \quad \mathbb{E}[G\mathbf{1}_{\{J=t\}}] = \frac{m_t}{2}.$$

Consequently $X = \mathbb{E}[G \mid J]$ satisfies $X \preceq_{\text{cx}} G$, and on every positive-probability event $\{J = t\}$,

$$X = \frac{m_t}{p_t + r_t}, \quad \|X\|_\infty = \frac{a_t}{p_t + r_t} \leq 1.$$

Apply Theorem 1.2 to the convex function $\Phi(x) = \max_{t \in T} \langle t, x \rangle$. We obtain

$$\frac{1}{2} \sum_{t \in T} \langle t, m_t \rangle = \mathbb{E} \langle J, X \rangle \leq \mathbb{E} \Phi(X) \leq C_0 b(T).$$

Taking the supremum in (4) yields $D(T) \leq 2C_0 b(T)$. The infimum in (3) is attained by coercivity. For a minimizer (v_t) set $U = \{t - v_t : t \in T\}$ and $V = \{v_t : t \in T\}$. Both terms in (3) are nonnegative. Hence

$$T \subset U + V, \quad \sup_{u \in U} \|u\|_1 \leq 2C_0 b(T), \quad g(V) \leq 2C_0 b(T),$$

as required. $\square$

3. PROOF OF THEOREM 1.2

The first lemma is the finite-support Gaussian case of a maximum-entropy martingale bridge. It is a special case of the construction of Hua, Song, and Tudose [4, Proposition 3.3 and Section 4.1], applied to $X/2 \preceq_{\text{cx}} G/2$. The general theory of martingale Schrödinger bridges is developed by Nutz and Wiesel [6] and by Backhoff, Beiglböck, Bifronte, and Ley [1]. We include the short argument for completeness.

Lemma 3.1. *Suppose X has finite support and $X \preceq_{\text{cx}} G$. There exists a coupling of X and $G \sim N(0, I_n)$ such that*

$$\mathbb{E}[G \mid X] = \frac{1}{2}X, \quad (5)$$

and each conditional distribution $G \mid \{X = x\}$ has an everywhere positive, 1-strongly log-concave density. Among couplings satisfying (5), this coupling uniquely maximizes $H(X \mid G)$.

Proof. Write $\mathbb{P}(X = x_j) = p_j > 0$ for $1 \leq j \leq m$. Convex comparison gives $\mathbb{E}X = 0$. Set $a_m = 0$, $b_m = 0$, and $\ell_j(g) = a_j + \langle b_j, g \rangle$. Consider

$$P(a, b) = \mathbb{E} \log \sum_{j=1}^m e^{\ell_j(G)} - \sum_{j=1}^m p_j \left(a_j + \frac{1}{2} \langle b_j, x_j \rangle \right). \quad (6)$$

Let $h = \mathbb{E} \max_j \ell_j(G)$. Since log-sum-exp dominates the maximum,

$$P(a, b) \geq \frac{1}{2} \left(h - \sum_j p_j \ell_j(x_j) \right) + \frac{1}{2} \mathbb{E} \left[\max_j \ell_j(G) - \sum_j p_j \ell_j(G) \right]. \quad (7)$$

The first term is nonnegative by $X \preceq_{\text{cx}} G$. The second is continuous, positively homogeneous, and strictly positive unless all ℓ_j coincide, hence unless $(a, b) = 0$. Compactness of the unit sphere now shows that P is coercive, so it has a minimizer (a^*, b^*) .

Define

$$q_j(g) = \frac{e^{a_j^* + \langle b_j^*, g \rangle}}{\sum_{k=1}^m e^{a_k^* + \langle b_k^*, g \rangle}}. \quad (8)$$

Differentiating (6) gives

$$\mathbb{E} q_j(G) = p_j, \quad \mathbb{E}[G q_j(G)] = \frac{1}{2} p_j x_j \quad (1 \leq j < m).$$

The identities for $j = m$ follow by summation from $\mathbb{E}G = \mathbb{E}X = 0$. Conditional on $G = g$, assign $X = x_j$ with probability $q_j(g)$. This has the prescribed X marginal and satisfies (5).

To see the maximum-entropy claim, let $\tilde{q}$ be any other kernel with the same marginal and barycenter constraints, and write $Z(g) = \sum_k e^{a_k^* + \langle b_k^*, g \rangle}$ and $H(\tilde{q}) = -\mathbb{E} \sum_j \tilde{q}_j(G) \log \tilde{q}_j(G)$. Since $\log q_j(g) = a_j^* + \langle b_j^*, g \rangle - \log Z(g)$, those constraints give

$$\mathbb{E} \sum_j \tilde{q}_j(G) \log q_j(G) = \sum_j p_j \left(a_j^* + \frac{1}{2} \langle b_j^*, x_j \rangle \right) - \mathbb{E} \log Z(G) = \mathbb{E} \sum_j q_j(G) \log q_j(G).$$

Thus the cross-entropy term is the same for every feasible kernel, and

$$H(q) - H(\tilde{q}) = \mathbb{E} \sum_j \tilde{q}_j(G) \log \frac{\tilde{q}_j(G)}{q_j(G)} \geq 0.$$

Equality holds only when $\tilde{q} = q$ almost surely. Finally, $\log q_j$ is concave because $\log$ -sum- $\exp$ is convex. Hence the conditional Gaussian density $g \mapsto [p_j(2\pi)^{n/2}]^{-1} q_j(g) e^{-\|g\|_2^2/2}$ is everywhere positive and 1-strongly log-concave. $\square$

The following is an immediate consequence of the Brascamp–Lieb variance inequality [3, Theorem 4.1].

Lemma 3.2. *For the coupling in Lemma 3.1,*

$$\text{Var}(G_i | X) \leq 1, \quad \mathbb{E}[G_i^2 | X] \leq 1 + \frac{1}{4} X_i^2.$$

We next record a standard subgaussian estimate needed for Lemma 3.4.

Lemma 3.3. *Let $V_1, \dots, V_n$ be independent and centered, with $\mathbb{E}e^{|V_i|/\kappa} \leq 2$, and let $Y_i = \mathbb{E}[V_i | \mathcal{H}]$ satisfy $|Y_i| \leq D$. Then*

$$\log \mathbb{E}e^{\langle a, Y \rangle} \leq \max\{4\kappa^2, 2\kappa D\} \|a\|_2^2 \quad (a \in \mathbb{R}^n).$$

Proof. The standard Bernstein moment estimate for a centered subexponential variable [10, proof of Proposition 2.8.1] gives

$$\mathbb{E}e^{sV_i} \leq e^{4\kappa^2 s^2} \quad \text{if } \kappa|s| \leq \frac{1}{2}.$$

Set $A = \{i : \kappa|a_i| > 1/2\}$. For $i \in A$, $D|a_i| \leq 2\kappa D a_i^2$. Thus conditional Jensen and independence give

$$\mathbb{E}e^{\langle a, Y \rangle} \leq e^{2\kappa D \sum_{i \in A} a_i^2} \prod_{i \notin A} \mathbb{E}e^{a_i V_i} \leq \exp \left(2\kappa D \sum_{i \in A} a_i^2 + 4\kappa^2 \sum_{i \notin A} a_i^2 \right).$$

The claimed bound follows. $\square$

Lemma 3.4. *Suppose $\|X\|_\infty \leq 1$, and use the coupling of Lemma 3.1. For every $L > 0$, there exist a uniform $\varepsilon \in \{-1, 1\}^n$ and a centered $Y \in \mathbb{R}^n$ such that*

$$\frac{1}{2}X = L\mathbb{E}[\varepsilon | X] + Y, \quad \|Y\|_\infty \leq \frac{1}{2L}, \quad \log \mathbb{E}e^{\langle a, Y \rangle} \leq \frac{4}{L^2} \|a\|_2^2. \quad (9)$$

Proof. Independently of (G, X) , take independent $U_i \sim \text{Unif}[-L, L]$ and set $\varepsilon_i = \text{sgn}(G_i + U_i)$. The pairs (G_i, U_i) are independent and symmetric. Therefore the ε_i are genuinely independent uniform signs; no independence conditional on X is asserted or needed. Moreover,

$$\mathbb{E}[\varepsilon_i | G, X] = \text{clip}(G_i/L, -1, 1).$$

Define $V_i = \text{sgn}(G_i)(|G_i| - L)_+$ and $Y_i = \mathbb{E}[V_i | X]$. The V_i are independent and centered, and $G_i = L \text{clip}(G_i/L, -1, 1) + V_i$. Taking conditional expectations and using (5) proves the first identity in (9).

The elementary inequality $(r - L)_+ \leq r^2/(4L)$ gives

$$\mathbb{E}e^{L|V_i|} \leq \mathbb{E}e^{G_i^2/4} = \sqrt{2} \leq 2.$$

Lemma 3.2 and $|X_i| \leq 1$ further give

$$|Y_i| \leq \frac{\mathbb{E}[G_i^2 | X]}{4L} \leq \frac{1 + X_i^2/4}{4L} \leq \frac{1}{2L}.$$

Apply Lemma 3.3 with $\kappa = L^{-1}$ and $D = (2L)^{-1}$ to obtain the final bound in (9). $\square$

Proof of Theorem 1.2. For a convex $\phi : \mathbb{R}^n \rightarrow \mathbb{R}$, write

$$S_\phi = \sup\{\mathbb{E}\phi(X) : X \preceq_{\text{cx}} G, \|X\|_\infty \leq 1\} < \infty.$$

It suffices to show that $S_\phi \leq \mathbb{E}\phi(C_0\varepsilon)$ for every convex ϕ . Conditional expectation on increasingly fine finite partitions of the cube shows that the supremum defining S_ϕ may be taken over finite-valued X .

Set $C_* = \max\{2\sqrt{2}C_G, 1/2\}$, fix $L > 2C_*$, and put $\lambda = C_*/L$ and $C_0 = 2L/(1 - 2\lambda)$. For each finite-valued admissible X , Lemma 3.4 and Gaussian comparison (2) yield a uniform sign vector ε and $G' \sim N(0, I_n)$ such that

$$\frac{1}{2}X = L\mathbb{E}[\varepsilon \mid X] + \lambda Z, \quad Z = \mathbb{E}[G' \mid X].$$

Jensen gives $Z \preceq_{\text{cx}} G$, and Lemma 3.4 gives

$$\|Z\|_\infty \leq \frac{1}{2L\lambda} = \frac{1}{2C_*} \leq 1, \quad X = (1 - 2\lambda)\mathbb{E}[C_0\varepsilon \mid X] + 2\lambda Z.$$

Thus another application of Jensen gives

$$\mathbb{E}\phi(X) \leq (1 - 2\lambda)\mathbb{E}\phi(C_0\varepsilon) + 2\lambda\mathbb{E}\phi(Z) \leq (1 - 2\lambda)\mathbb{E}\phi(C_0\varepsilon) + 2\lambda S_\phi.$$

Taking the supremum over X and using $2\lambda < 1$ yields $S_\phi \leq \mathbb{E}\phi(C_0\varepsilon)$, as claimed. $\square$

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